Evaluate Expression Tree

AlgoExpert · Easy · Binary Trees

The problem

- Leaves are positive ints; internal nodes encode operators -1 '+', -2 '-', -3 '/' (truncating), -4 '*'.
- Evaluate the tree.

Intuition

- Leaves are numbers and internal nodes are operators, so evaluate the children first and combine them.

Approach

- 1. If the node is a leaf, return its value
- 2. Evaluate left and right subtrees
- 3. Apply the operator encoded at the node
- 4. Division truncates toward zero

Java solution

```java
public class EvaluateExpressionTree {
    public static int evaluateExpressionTree(BinaryTree t) {
        if (t.value >= 0) return t.value;
        int l = evaluateExpressionTree(t.left), r = evaluateExpressionTree(t.right);
        return switch (t.value) {
            case -1 -> l + r;
            case -2 -> l - r;
            case -3 -> l / r;
            default -> l * r;
        };
    }
}
```

Complexity

- Time: O(n)
- Space: O(h)

Recap

- Leaves are numbers and internal nodes are operators, so evaluate the children first and combine them.
- Complexity: O(n)